TP15 : Packager une application Kubernetes avec Helm et Kustomize

Installer Helm :

```
arij@vagrant:~/DevOps_tp15$ curl -fsSL https://raw.githubusercontent.com/helm/helm/main/scripts/get-helm-3 | bash
helm version
Downloading https://get.helm.sh/helm-v3.22.1-linux-amd64.tar.gz
Verifying checksum... Done.
Preparing to install helm into /usr/local/bin
[sudo] password for arij:
helm installed into /usr/local/bin/helm
version.BuildInfo{Version:"v3.22.1", GitCommit:"d6c1a0e4027caa6abc7892e7ea9f0cda7447baa3", GitTreeState:"clean", GoVersion:"go1.26.9"
}
```

Partie 1 : Consommer un chart public

1. Ajouter le dépôt et chercher

```
arij@vagrant:~/DevOps_tp15$ helm repo add podinfo https://stefanprodan.github.io/podinfo
"podinfo" has been added to your repositories
arij@vagrant:~/DevOps_tp15$ helm repo update
Hang tight while we grab the latest from your chart repositories...
...Successfully got an update from the "podinfo" chart repository
Update Complete. ⎈Happy Helming!⎈
arij@vagrant:~/DevOps_tp15$ helm search repo podinfo
NAME            CHART VERSION   APP VERSION     DESCRIPTION
podinfo/podinfo 6.17.0          6.17.0          Podinfo Helm chart for Kubernetes
arij@vagrant:~/DevOps_tp15$
```

2. Inspecter le chart :

```
arij@vagrant:~/DevOps_tp15$ helm show chart podinfo/podinfo
apiVersion: v1
appVersion: 6.17.0
description: Podinfo Helm chart for Kubernetes
home: https://github.com/stefanprodan/podinfo
kubeVersion: '>=1.23.0-0'
maintainers:
- email: stefanprodan@users.noreply.github.com
  name: stefanprodan
name: podinfo
sources:
- https://github.com/stefanprodan/podinfo
version: 6.17.0

arij@vagrant:~/DevOps_tp15$ helm show values podinfo/podinfo | less
arij@vagrant:~/DevOps_tp15$
```

3.Installer avec une surcharge :

```
arij@vagrant:~/DevOps_tp15$ helm install demo podinfo/podinfo -n helm-demo --create-namespace --set replicaCount=2
NAME: demo
LAST DEPLOYED: Fri Nov 13 12:44:59 2026
NAMESPACE: helm-demo
STATUS: deployed
REVISION: 1
NOTES:
1. Get the application URL by running these commands:
  echo "Visit http://127.0.0.1:8080 to use your application"
  kubectl -n helm-demo port-forward deploy/demo-podinfo 8080:9898
arij@vagrant:~/DevOps_tp15$
```

4.Vérifier l'installation et l'historique **:**

```
arij@vagrant:~/DevOps_tp15$ helm list -A
NAME            NAMESPACE       REVISION        UPDATED                                 STATUS          CHART
APP VERSION
demo            helm-demo       1               2026-11-13 12:44:59.071259088 +0000 UTC deployed        podinfo-6.17.0
6.17.0
gateway-api-crd kube-system     1               2026-11-12 20:16:53.620731297 +0000 UTC deployed        gateway-api-crd-1.6.301
v1.6.3
traefik         kube-system     1               2026-11-12 20:17:13.40496189 +0000 UTC  deployed        traefik-41.6.101+up41.6.1
v3.7.13
traefik-crd     kube-system     1               2026-11-12 20:16:54.283093647 +0000 UTC deployed        traefik-crd-41.6.101+up41.6.1
v3.7.1
arij@vagrant:~/DevOps_tp15$ helm get values demo -n helm-demo
USER-SUPPLIED VALUES:
replicaCount: 2
arij@vagrant:~/DevOps_tp15$ helm get manifest demo -n helm-demo | head -40
---
# Source: podinfo/templates/service.yaml
apiVersion: v1
kind: Service
metadata:
  name: demo-podinfo
  namespace: helm-demo
  labels:
    helm.sh/chart: podinfo-6.17.0
    app.kubernetes.io/name: demo-podinfo
    app.kubernetes.io/version: "6.17.0"
    app.kubernetes.io/managed-by: Helm
spec:
  type: ClusterIP
  ports:
    - port: 9898
      targetPort: http
      protocol: TCP
```

```
arij@vagrant:~/DevOps_tp15$ kubectl -n helm-demo get secrets
NAME                          TYPE                 DATA   AGE
sh.helm.release.v1.demo.v1    helm.sh/release.v1   1      103s
arij@vagrant:~/DevOps_tp15$
```

5. Désinstaller **:**

```
arij@vagrant:~/DevOps_tp15$ helm uninstall demo -n helm-demo
release "demo" uninstalled
arij@vagrant:~/DevOps_tp15$
```

Partie 2 : Écrire son propre chart

```
arij@vagrant:~/DevOps_tp15$ nano chart-starter/whoami/templates/deployment.yaml
arij@vagrant:~/DevOps_tp15$ nano chart-starter/whoami/templates/service.yaml
arij@vagrant:~/DevOps_tp15$ nano chart-starter/whoami/templates/ingress.yaml
arij@vagrant:~/DevOps_tp15$ helm lint chart-starter/whoami
==> Linting chart-starter/whoami
[INFO] Chart.yaml: icon is recommended

1 chart(s) linted, 0 chart(s) failed
arij@vagrant:~/DevOps_tp15$ helm template test chart-starter/whoami --set ingress.enabled=true | less
```

```
arij@vagrant:~/DevOps_tp15$ helm install test chart-starter/whoami --dry-run=server -n tp15 --create-namespace
NAME: test
LAST DEPLOYED: Fri Nov 13 12:51:26 2026
NAMESPACE: tp15
STATUS: pending-install
REVISION: 1
TEST SUITE: None
HOOKS:
MANIFEST:
---
# Source: whoami/templates/service.yaml
apiVersion: v1
kind: Service
metadata:
  name: test-whoami
  labels:
    app.kubernetes.io/name: whoami
    app.kubernetes.io/instance: test
    app.kubernetes.io/version: "v1.10.0"
    app.kubernetes.io/managed-by: Helm
    helm.sh/chart: whoami-0.1.0
spec:
  type: ClusterIP
  ports:
    - port: 80
      targetPort: http
      protocol: TCP
      name: http
  selector:
    app.kubernetes.io/name: whoami
    app.kubernetes.io/instance: test
---
# Source: whoami/templates/deployment.yaml
```

Partie 3 : Cycle de vie d'une release

1. Installer la release avec l'environnement "dev"

```
arij@vagrant:~/DevOps_tp15$ helm install whoami chart-starter/whoami -n tp15 --create-namespace -f values-dev.yaml
NAME: whoami
LAST DEPLOYED: Fri Nov 13 13:11:45 2026
NAMESPACE: tp15
STATUS: deployed
REVISION: 1
TEST SUITE: None
NOTES:
Release whoami installée (révision 1).
Application : http://whoami.lab.local
arij@vagrant:~/DevOps_tp15$ kubectl -n tp15 get pods
NAME                              READY   STATUS    RESTARTS   AGE
whoami-whoami-766bdcb8d8-kgqsc   1/1     Running   0          26s
arij@vagrant:~/DevOps_tp15$ curl http://whoami.lab.local/?env=true
Name: dev
Hostname: whoami-whoami-766bdcb8d8-kgqsc
IP: 127.0.0.1
IP: ::1
IP: 10.42.0.47
IP: fe80::838:a2ff:fe8e:de97
RemoteAddr: 10.42.0.18:37420
GET /?env=true HTTP/1.1
Host: whoami.lab.local
User-Agent: curl/8.5.0
Accept: */*
Accept-Encoding: gzip
X-Forwarded-For: 10.42.0.1
X-Forwarded-Host: whoami.lab.local
X-Forwarded-Port: 80
X-Forwarded-Proto: http
X-Forwarded-Server: traefik-68df944ff7-fqz4g
X-Real-Ip: 10.42.0.1
```

2.Faire une mise à jour (Upgrade) valide

```
arij@vagrant:~/DevOps_tp15$ helm upgrade whoami chart-starter/whoami -n tp15 -f values-dev.yaml --set replicaCount=3
Release "whoami" has been upgraded. Happy Helming!
NAME: whoami
LAST DEPLOYED: Fri Nov 13 13:13:00 2026
NAMESPACE: tp15
STATUS: deployed
REVISION: 2
TEST SUITE: None
NOTES:
Release whoami installée (révision 2).
Application : http://whoami.lab.local
arij@vagrant:~/DevOps_tp15$
```

3.Provoquer un échec (Image inexistante) et faire un Rollback

```
arij@vagrant:~/DevOps_tp15$ helm upgrade whoami chart-starter/whoami -n tp15 -f values-dev.yaml --set image.tag=v9.9.9
Release "whoami" has been upgraded. Happy Helming!
NAME: whoami
LAST DEPLOYED: Fri Nov 13 13:13:37 2026
NAMESPACE: tp15
STATUS: deployed
REVISION: 3
TEST SUITE: None
NOTES:
Release whoami installée (révision 3).
Application : http://whoami.lab.local
arij@vagrant:~/DevOps_tp15$ kubectl -n tp15 get pods
NAME                      READY   STATUS             RESTARTS   AGE
whoami-whoami-766bdcb8d8-kgqsc   1/1     Running            0          2m11s
whoami-whoami-7d9667f969-hdjbr   0/1     ImagePullBackOff   0          19s
arij@vagrant:~/DevOps_tp15$ helm history whoami -n tp15
REVISION        UPDATED                         STATUS          CHART           APP VERSION     DESCRIPTION
1               Fri Nov 13 13:11:45 2026        superseded      whoami-0.1.0    v1.10.0         Install complete
2               Fri Nov 13 13:13:00 2026        superseded      whoami-0.1.0    v1.10.0         Upgrade complete
3               Fri Nov 13 13:13:37 2026        deployed        whoami-0.1.0    v1.10.0         Upgrade complete
arij@vagrant:~/DevOps_tp15$ helm rollback whoami 2 -n tp15
Rollback was a success! Happy Helming!
arij@vagrant:~/DevOps_tp15$ helm history whoami -n tp15
REVISION        UPDATED                         STATUS          CHART           APP VERSION     DESCRIPTION
1               Fri Nov 13 13:11:45 2026        superseded      whoami-0.1.0    v1.10.0         Install complete
2               Fri Nov 13 13:13:00 2026        superseded      whoami-0.1.0    v1.10.0         Upgrade complete
3               Fri Nov 13 13:13:37 2026        superseded      whoami-0.1.0    v1.10.0         Upgrade complete
4               Fri Nov 13 13:14:13 2026        deployed        whoami-0.1.0    v1.10.0         Rollback to 2
arij@vagrant:~/DevOps_tp15$
```

4.Tester l'option --atomic (Ta capture helm-atomic.png)

```
arij@vagrant:~/DevOps_tp15$ helm upgrade whoami chart-starter/whoami -n tp15 -f values-dev.yaml --set image.tag=v9.9.9 --atomic --tim
eout 60s
Error: UPGRADE FAILED: release whoami failed, and has been rolled back due to atomic being set: context deadline exceeded
arij@vagrant:~/DevOps_tp15$
```

5. Packager le chart :

```
arij@vagrant:~/DevOps_tp15$ helm package chart-starter/whoami
Successfully packaged chart and saved it to: /home/arij/DevOps_tp15/whoami-0.1.0.tgz
arij@vagrant:~/DevOps_tp15$
```

Partie 4 : Kustomize, base et overlays

1.Prévisualiser la base et l'overlay dev

```
arij@vagrant:~/DevOps_tp15$ kubectl kustomize kustomize-starter/base
apiVersion: v1
data:
  LOG_LEVEL: info
  WHOAMI_NAME: base
kind: ConfigMap
metadata:
  labels:
    app.kubernetes.io/part-of: tp15
  name: whoami-config-29777g9m9k
---
apiVersion: v1
kind: Service
metadata:
  labels:
    app.kubernetes.io/part-of: tp15
  name: whoami
spec:
  ports:
  - name: http
    port: 80
    targetPort: http
  selector:
    app: whoami
---
apiVersion: apps/v1
kind: Deployment
metadata:
  labels:
    app.kubernetes.io/part-of: tp15
  name: whoami
spec:
  replicas: 1
```

```
arij@vagrant:~/DevOps_tp15$ kubectl kustomize kustomize-starter/overlays/dev
apiVersion: v1
kind: Namespace
metadata:
  name: tp15-dev
---
apiVersion: v1
data:
  LOG_LEVEL: debug
  WHOAMI_NAME: dev
kind: ConfigMap
metadata:
  labels:
    app.kubernetes.io/part-of: tp15
  name: whoami-config-6t5d5gdccm
  namespace: tp15-dev
---
apiVersion: v1
kind: Service
metadata:
  labels:
    app.kubernetes.io/part-of: tp15
  name: whoami
  namespace: tp15-dev
spec:
  ports:
  - name: http
    port: 80
```

3.Déployer les trois environnements

```
arij@vagrant:~/DevOps_tp15$ kubectl apply -k kustomize-starter/overlays/dev
namespace/tp15-dev created
configmap/whoami-config-6t5d5gdccm created
service/whoami created
deployment.apps/whoami created
arij@vagrant:~/DevOps_tp15$ kubectl apply -k kustomize-starter/overlays/staging
configmap/whoami-config-fmf7t566f2 created
service/whoami created
deployment.apps/whoami created
arij@vagrant:~/DevOps_tp15$ kubectl apply -k kustomize-starter/overlays/prod
configmap/whoami-config-29777g9m9k created
service/whoami unchanged
deployment.apps/whoami configured
arij@vagrant:~/DevOps_tp15$
```

4.Vérifier les trois déploiements

```
arij@vagrant:~/DevOps_tp15$ kubectl get deploy -A -l app.kubernetes.io/part-of=tp15
NAMESPACE   NAME     READY   UP-TO-DATE   AVAILABLE   AGE
default     whoami   3/3     3            3           29s
tp15-dev    whoami   1/1     1            1           33s
arij@vagrant:~/DevOps_tp15$
```

5. Comparer staging et prod :

```
arij@vagrant:~/DevOps_tp15$ diff <(kubectl kustomize kustomize-starter/overlays/staging) <(kubectl kustomize kustomize-starter/overla
ys/prod)
4c4
<   WHOAMI_NAME: staging
---
>   WHOAMI_NAME: base
9c9
<   name: whoami-config-fmf7t566f2
---
>   name: whoami-config-29777g9m9k
32c32
<   replicas: 2
---
>   replicas: 3
47,48c47,48
<             name: whoami-config-fmf7t566f2
<         image: traefik/whoami:v1.10.0
---
>             name: whoami-config-29777g9m9k
>         image: traefik/whoami:v1.10.1
52a53,58
>         readinessProbe:
>           httpGet:
>             path: /health
>             port: http
>           initialDelaySeconds: 5
>           periodSeconds: 5
55c61
<             memory: 64Mi
---
>             memory: 128Mi
arij@vagrant:~/DevOps_tp15$
```